Cryptanalysis of Multi-Party Key Exchange Protocols over a Modified Supertropical Semiring

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Abstract

We present a cryptanalysis of a multi-party key exchange protocol over a modified supertropical semiring, as proposed in a recent work of R. Ponmaheshkumar, J. Ramalingam, and R. Perumal. Building on the established methods for solving linear systems $A \otimes x = b$ over the tropical semiring, as well as on our recent work on solving such systems over layered semirings such as the symmetrized and supertropical semirings, we develop a method to compute a solution of $A \otimes x = b$ over the above mentioned modified supertropical semiring. This method enables the attacker to recover the shared secret key by solving the one-sided linear system derived from the public messages of the protocol. Our findings show that this modified supertropical platform does not provide the intended security and motivate further exploration of secure semiring-based constructions.

Keywords: public key cryptography; key exchange protocol; cryptographic attack; tropical cryptography; linear systems of equations; supertropical semiring
Classification: 94A60, 15A80

1 Introduction

Semirings are algebraic structures that generalize rings but lack additive inverses in general. The tropical (max-plus) semiring is arguably one of the most intensely studied and has applications in discrete event systems, railway timetabling, and network optimization. Its applications have also recently been extended to the field of cryptography. Algebraic cryptography has some uses in cryptographic key exchange protocols by which two users exchange messages to derive a common secret key to which an eavesdropper has no direct access. These protocols rely on the computational difficulty of dealing with some mathematical problems that arise in the domains of number theory, algebraic geometry, and combinatorics in order to give security against unauthorized access. For instance, the RSA scheme, among the earliest and most widely known, relies on the hardness of factoring large composite numbers into prime ones. As computational power increases, particularly with quantum computing challenging many traditional security assumptions, there is considerable interest in exploring new algebraic frameworks that could support secure and efficient cryptosystems. Semirings, and in particular the tropical semiring, have emerged as a promising direction. Tropical cryptography seeks to utilize structures from tropical mathematics to adapt classical public-key exchange protocols, such as those proposed by Diffie and Hellman [5] and Stickel [11].

Grigoriev and Shpilrain were among the first to incorporate tropical algebra as an alternative platform for cryptographic protocols [6]. They developed a tropical version of the Stickel key exchange to replace the original, which had been vulnerable to standard linear algebra attacks. This was driven by the general non-invertibility of matrices in tropical algebra, which could resist straightforward analogs of those attacks, along with the efficiency of tropical operations. However, this tropical implementation was later found to be insecure, as one-sided tropical linear systems $A \otimes x = b$ can be solved relatively easily. The recent attack by Otero-Sanchez et al. [9], which we are slightly generalizing in Section 4, shows how to exploit this weakness. This has motivated investigations into other implementations of the Stickel protocol over similar semirings. One approach involves layered tropical semirings, such as the supertropical semiring introduced by Izhakian [7], which distinguishes between tangible and ghost elements and uses the ghost surpass relation to recover certain features of classical linear algebra. This can be viewed as a special case of the tropical layered semirings discussed by Akian, Gaubert, and Guterman [2]. Linear algebra over a more general structure known as semiring pairs has been developed more recently by Akian, Gaubert, and Rowen [1]. This idea to explore other extensions of the tropical semiring has been one of the motivations for Ponmaheshkumar, Ramalingam and Perumal [10] who suggested a modified supertropical semiring as a platform for a multi-party key exchange protocol (being their original generalization of the Stickel protocol). The objective of this paper is to present a method, extending our recent work on describing the solution set for linear systems $A \otimes x = b$ over layered semirings such as the symmetrized and supertropical semirings [3], for finding a solution to one-sided linear systems $A \otimes x = b$ over the above mentioned modified supertropical semiring (which we prefer to call lex-supertropical below) and showing how this solution can be leveraged to break the suggested key exchange protocol.

This paper is organized as follows: Section 2 provides the necessary preliminaries, introduces the lex-supertropical semiring proposed in [10], and presents the relevant definitions. Section 3 describes and proves a method for finding a solution to one-sided linear systems over the lex-supertropical semiring. Finally, Section 4 demonstrates how the results from Section 3 can be used to attack the proposed multi-party key exchange protocol and presents the corresponding numerical experiments.

2 Preliminaries

This section presents the lex-supertropical semiring introduced in [10] and the associated matrix algebra. Throughout, we use the conventional notation $[m] = \{1, \dots, m\}$ and $[n] = \{1, \dots, n\}$ for most common index sets.

Definition 2.1 (Semiring)**.** Let S be a non-empty set equipped with two binary operations $\oplus$ and $\otimes$, which satisfy the following properties:

- $(S, \oplus)$ is a commutative monoid which means that it satisfies associativity, commutativity and existence of an additive identity element ϵ.

- $(S, \otimes)$ is a monoid which means that it satisfies associativity and existence of multiplicative identity element e.

- In $(S, \oplus, \otimes)$ multiplication $\otimes$ distributes over addition $\oplus$.
- The additive identity ϵ satisfies the absorbing property, that is $\epsilon \otimes e = e \otimes \epsilon = \epsilon$.

By extending the arithmetic operations of a semiring to matrices and vectors in a usual way, one defines the linear algebra over this semiring and, furthermore, poses a question of how difficult it is to solve linear systems of the form $A \otimes z = b$ over it (with A being a matrix and z and b being vectors of appropriate dimensions).

Definition 2.2 (Lex-tropical semiring)**.** The lex-tropical semiring $\mathbb{Z}^p_{\max}$ is defined as the set $\mathbb{Z}^p_{\max} = (\mathbb{Z}^p \cup \{-\infty\}, \oplus, \otimes)$, where the tropical addition $\oplus$ is defined by $a \oplus b = \max(a, b)$ for all $a, b \in \mathbb{Z}^p_{\max}$ where the maximum is taken with respect to the usual lexicographic order $\leq$ on the integer tuples, and the tropical multiplication $\otimes$ is defined by $a \otimes b = a + b$ for all $a, b \in \mathbb{Z}^p_{\max}$ where the addition is taken componentwise. In what follows, we will often use the usual arithmetic notation $a + b$ for this tropical multiplication (meaning the usual addition of the integer tuples unless one of them is the absorbing element $-\infty$), as well as $a - b$ for the subtraction of integer tuples (assuming that $b \neq -\infty$).

This semiring is very similar to the usual tropical semiring that can be defined over $\mathbb{Z} \cup \{-\infty\}$ with arithmetics $a \oplus b = \max(a, b)$ and $a \otimes b = a + b$. In particular the following result holds for the linear systems $A \otimes z = b$ over this semiring. We include the proof in Appendix, but it is a standard result whose proof is similar to that of e.g. Butkovič [4] Theorem 3.1.1. (or can be deduced from more recent results of Otero-Sanchez et al. [9]).

Proposition 2.1 (e.g. [4],[9])**.** Let $A \in (\mathbb{Z}^p_{\max})^{m \times n}$ have a finite entry in each column and $b \in (\mathbb{Z}^p)^m$ and define $\bar{z}$ by

$$\bar{z}_j = \min_{i\colon A_{ij} \neq -\infty} b_i - A_{ij}. \tag{1}$$

Then, the system $A \otimes z = b$ is solvable if and only if $A \otimes \bar{z} = b$, in which case $\bar{z}$ is the greatest solution.

Proof. See Appendix. □

Following [10] we now define the supertropical semiring based on $\mathbb{Z}^p_{\max}$.

Remark 2.1. The construction in [10] is more general than what we are going to present, as it uses a general multiplicatively cancellative totally ordered idempotent semiring instead of $\mathbb{Z}^p_{\max}$. It can be argued, however, that the Stickel protocol can be implemented over any semiring and in its cryptanalysis one needs to focus on a more concrete numerical implementation (with appropriate sacrifice of generality). To this end, the lex-supertropical semiring which we introduce here is a generalisation of the supertropical semiring of integer triples (see Example 2.1 below) over which the Stickel protocol is actually implemented in [10].

Definition 2.3 (Underlying set and layers)**.** Let

$$T = \mathbb{Z}^p \ \cup\ (\mathbb{Z}^p)^\circ \ \cup\ \{-\infty\},$$

where $\mathbb{Z}^p$ is the *tangible* layer (integer tuples with p elements), $(\mathbb{Z}^p)^\circ$ is a disjoint *ghost* copy of $\mathbb{Z}^p$, and $-\infty$ plays the role of the additive identity and multiplicatively absorbing element.

Definition 2.4 (Projection)**.** Consider any homomorphism $g : \mathbb{Z}^p \to (\mathbb{Z}^p)^\circ$, i.e., a mapping between these lattices satisfying the property $g(a+b) = g(a) + g(b)$.

Define the projection $f : T \to (\mathbb{Z}^p)^\circ \cup \{-\infty\}$ by

$$f(x) = \begin{cases} g(x), & \text{if } x \in \mathbb{Z}^p, \\ x, & \text{if } x \in (\mathbb{Z}^p)^\circ \cup \{-\infty\}. \end{cases}$$

The addition on T uses the lexicographic order on $(\mathbb{Z}^p)^\circ \cup \{-\infty\}$ naturally assuming that $-\infty$ is the smallest element with respect to this order.

Example 2.1. In [10], the lex-supertropical semiring utilized to implement the Stickel protocol has $p = 3$. In this case, the lexicographic order defined on integer triples (p_1, q_1, r_1) and (p_2, q_2, r_2) works as follows: $(p_1, q_1, r_1) > (p_2, q_2, r_2)$ if $p_1 > p_2$ or $p_1 = p_2$ and $q_1 > q_2$ or $p_1 = p_2$, $q_1 = q_2$ and $r_1 > r_2$, and $(p_1, q_1, r_1) < (p_2, q_2, r_2)$ otherwise.

The following function is used for g:

$$(p, q, r) \in \mathbb{Z}^3 \mapsto (a, b, c) = (p + q + r,\ p - q + r,\ p + q - r). \tag{2}$$

Function (2) happens to be injective, and it maps $\mathbb{Z}^3$ injectively onto a proper sublattice of $\mathbb{Z}^3$ consisting precisely of those triples (a, b, c) for which $a - b$ and $a - c$ are even. However, such properties of g will not be important for our attack, as the rest of the paper will show.

Definition 2.5 (Addition)**.** For $a, b \in T$ define $a \oplus b$ by comparing $f(a), f(b)$:

$$a \oplus b = \begin{cases} a, & f(a) > f(b), \\ b, & f(a) < f(b), \\ f(a), & f(a) = f(b). \end{cases}$$

Definition 2.6 (Multiplication)**.** Multiplication is componentwise addition of the underlying integer tuples, with the following layer rules:

- $-\infty \otimes c = c \otimes -\infty = -\infty$ for all $c \in T$.
- If $a, b \in \mathbb{Z}^p$ or $a, b \in (\mathbb{Z}^p)^\circ$ then $a \otimes b = a + b$.
- If $a \in \mathbb{Z}^p$ and $b \in (\mathbb{Z}^p)^\circ$ then $a \otimes b = f(a) + b (= g(a) + b)$.

The fact that the above construction forms a semiring was proved by Ponmaheshkumar et al. [10], but we include our original proof in Appendix for reader's convenience.

Proposition-Definition 2.1 (Lex-supertropical semiring [10])**.** The above construction ensures $(T, \oplus, \otimes)$ forms a semiring, which we call the lex-supertropical semiring.

Proof. See Appendix. □

The proof, which we give in Appendix, relies on the observation that f is linear (shown below).

Proposition 2.2. For any $a, b \in T$ we have: $f(a \otimes b) = f(a) + f(b)$, $f(a \oplus b) = f(a) \oplus f(b)$.

Proof. For $f(a \otimes b) = f(a) + f(b)$, in the case where $a, b \in \mathbb{Z}^p$ this follows since g is a homomorphism, and if $a, b \in (\mathbb{Z}^p)^\circ$ then it is trivial since $f(a) = a$ and $f(b) = b$ in this case, and the property is also obvious if $a = -\infty$ or $b = -\infty$. In the case where $a \in \mathbb{Z}^p$ and $b \in (\mathbb{Z}^p)^\circ$ then $f(a \otimes b) = f(a) + b = f(a) + f(b)$.

For $f(a \oplus b) = f(a) \oplus f(b)$, we consider two special cases: $f(a) \neq f(b)$ and $f(a) = f(b)$. In the first case by Definition 2.5 $f(a \oplus b)$ is the lexicographic maximum of $f(a)$ and $f(b)$, while in the second case it is $f(a) = f(a) \oplus f(a)$. $\square$

We immediately obtain the following corollaries of this linearity. Here and below, $f(A)$, $f(b)$ and $f(x)$ denote the matrix and the vectors whose entries are obtained as projections (under f) of the entries of A, b and x, respectively.

Corollary 1. *If x is a solution of $A \otimes x = b$ then $f(x)$ is a solution to $f(A) \otimes z = f(b)$.*

Proof. Apply f to both sides of each equation of $A \otimes x = b$ and use the linearity of f established in Proposition 2.2. $\square$

The following fact is then an easy combination of Proposition 2.1 and the linearity of f.

Corollary 2. *Let $A \in T^{m \times n}$ have a finite entry in each column and $b \in (T \backslash \{-\infty\})^m$, and let $x \in T^n$ satisfy $A \otimes x = b$. Define $\bar{z}$ by*

$$\bar{z}_j = \min_{i\colon A_{ij} \neq -\infty} f(b_i) - f(A_{ij}). \tag{3}$$

Then, $f(x) \leq \bar{z}$.

Remark 2.2. In the above results on $A \otimes x = b$ over lex-tropical and lex-supertropical semirings we assumed that b has only finite entries and A has a finite entry in each column. For solving $A \otimes x = b$ over such semirings these assumptions are not restrictive as a system not satisfying such assumptions can be reduced to a system satisfying them. Indeed, for each $b_i = -\infty$ we search for j with $A_{ij} \neq -\infty$ and then, knowing that x_j should be $-\infty$ we can delete the entire column for every such j. The remaining system will have only finite b_i on the right-hand side and it is solvable if and only if the initial system is solvable. Next, whenever there is a column j of A without finite entries we can also delete it since none of the terms $A_{ij} \otimes x_j$ is contributing to the sums on the left-hand side and x_j can be taken as arbitrary semiring element.

The operations of any semiring – including $\mathbb{Z}^p_{\max}$ and T – naturally extend to vectors and matrices, resulting in a corresponding matrix algebra. Let us in particular recall the following standard constructions which are used in the tropical Stickel protocol of Grigoriev and Shpilrain [6] (with its straightforward semiring generalization in mind).

Definition 2.7 (Matrix powers)**.** For $A \in T^{n \times n}$, the n-th power of A is denoted by $A^{\otimes n}$, and is equal to

$$A^{\otimes n} = \underbrace{A \otimes A \otimes \ldots \otimes A}_{n \text{ times}}$$

By definition, any square matrix to the power of zero equals the identity matrix.

Definition 2.8 (Identity matrix)**.** The identity matrix $I \in T^{n \times n}$ is of the form $(I)_{ij} = \delta_{ij}$ where

$$\delta_{ij} = \begin{cases} (0, \ldots, 0) & \text{if } i = j \\ -\infty & \text{otherwise} \end{cases}$$

We subsequently define the matrix polynomials over this semiring.

Definition 2.9 (Matrix polynomials)**.** Matrix polynomial is a function of the form

$$A \mapsto p(A) = \bigoplus_{k=0}^{d} a_k \otimes A^{\otimes k},$$

where $A \in T^{n \times n}$, and $a_k \in T$.

3 Finding one solution of $A \otimes x = b$ over the lex-supertropical semiring

In this section we explain how to find one solution of $A \otimes x = b$ over the lex-supertropical semiring. This task is relevant because finding such solution is sufficient for attacking certain cryptographic protocols over this semiring (as will be discussed in the upcoming section).

As a preliminary, we present the concept of exact covers and the corresponding enumeration algorithm, which will be utilized as an intermediate step in the proposed algorithms.

Definition 3.1 (Exact cover)**.** Let U be a finite set and $\mathcal{S} = \{S_1, S_2, \ldots, S_n\}$ be a collection of subsets of U. A subcollection $\mathcal{K} \subseteq \mathcal{S}$ is called an exact cover of U if each element of U belongs to exactly one subset in $\mathcal{K}$, i.e., $\bigcup_{S \in \mathcal{K}} S = U$ and $S_i \cap S_j = \emptyset$ for all distinct $S_i, S_j \in \mathcal{K}$.

The following algorithm, known as Algorithm X by Knuth [8], is a simple recursive backtracking method for enumerating exact covers. Although more efficient implementations exist, most notably Knuth's Dancing Links (DLX) [8], we use the basic version here since it already provides sufficient performance for the attack presented in the next section. However, employing a more optimized implementation could further improve its efficiency.

Algorithm 1 Enumerating exact covers

Require: A universe of elements U, and a family of subsets $\mathcal{S}$
Ensure: All exact covers $K \subseteq \mathcal{S}$ of U

procedure EnumerateExactCovers($U, \mathcal{S}, K$)
    if U is empty **then**
        Output K
        return
    Let c be an element of U contained in the minimal number of sets from $\mathcal{S}$
    for each set $R \in \mathcal{S}$ containing c **do**
        Add R to K
        Let V be the union of elements in R
        Remove V from U and remove from $\mathcal{S}$ all sets intersecting V
        EnumerateExactCovers($U, \mathcal{S}, K$) on reduced sets
        Restore U and $\mathcal{S}$
        Remove R from K

The toy example below illustrates how this algorithm works.

Example 3.1. Consider the universe $U = \{1,2,3\}$ and the family of subsets $\mathcal{S} = \{S_1 = \{1,2\}, S_2 = \{2\}, S_3 = \{3\}, S_4 = \{1,3\}\}$.

The algorithm starts with the initial call: EnumerateExactCovers($U = \{1,2,3\}$, $\mathcal{S} = \{S_1, S_2, S_3, S_4\}$, $K = \emptyset$).

1. U is not empty. Compute the number of sets containing each element:
   - 1 appears in 2 sets (S_1, S_4).
   - 2 appears in 2 sets (S_1, S_2).
   - 3 appears in 2 sets (S_3, S_4).

   All have the same count, so select $c = 1$ (arbitrarily, as any would work).

2. Iterate over sets in $\mathcal{S}$ containing $c = 1$: S_1 and S_4.
   - **First branch: Try** $R = S_1 = \{1,2\}$.
     - Add S_1 to K: $K = \{S_1\}$.
     - Let $V = \{1,2\}$ (the elements of R).
     - Remove V from U: $U = \{1,2,3\} \setminus \{1,2\} = \{3\}$.
     - Remove from $\mathcal{S}$ all sets intersecting $V = \{1,2\}$:
       * S_1 intersects (at 1 and 2) $\rightarrow$ remove.
       * S_2 intersects (at 2) $\rightarrow$ remove.
       * S_3 does not intersect $\rightarrow$ keep.
       * S_4 intersects (at 1) $\rightarrow$ remove.

     Reduced $\mathcal{S} = \{S_3\}$.

- Recurse: $\textsc{EnumerateExactCovers}(U = \{3\}, \mathcal{S} = \{S_3\}, K = \{S_1\})$.
  * In recursion: U is not empty. Select $c = 3$ (only element, it appears in 1 set: S_3).
  * Iterate over sets containing $c = 3$: only S_3.
  * Try $R = S_3 = \{3\}$.
    · Add S_3 to K: $K = \{S_1, S_3\}$.
    · Let $V = \{3\}$.
    · Remove V from U: $U = \{3\} \setminus \{3\} = \emptyset$.
    · Remove from $\mathcal{S}$ all sets intersecting $V = \{3\}$:
      ◦ S_3 intersects (at 3) $\rightarrow$ remove.

      Reduced $\mathcal{S} = \emptyset$.
    · Recurse: $\textsc{EnumerateExactCovers}(U = \emptyset, \mathcal{S} = \emptyset, K = \{S_1, S_3\})$.
      ◦ U is empty $\rightarrow$ Output $K = \{S_1, S_3\}$ (covers $\{1, 2, 3\}$ exactly).
    · Backtrack: Restore U to $\{3\}$ and $\mathcal{S}$ to $\{S_3\}$, remove S_3 from K: $K = \{S_1\}$.
  * No more sets containing $c = 3$. End recursion.
- Backtrack: Restore U to $\{1, 2, 3\}$ and $\mathcal{S}$ to $\{S_1, S_2, S_3, S_4\}$, remove S_1 from K: $K = \emptyset$.

• **Second branch: Try** $R = S_4 = \{1, 3\}$.
  - Add S_4 to K: $K = \{S_4\}$.
  - Let $V = \{1, 3\}$.
  - Remove V from U: $U = \{1, 2, 3\} \setminus \{1, 3\} = \{2\}$.
  - Remove from $\mathcal{S}$ all sets intersecting $V = \{1, 3\}$:
    * S_1 intersects (at 1) $\rightarrow$ remove.
    * S_2 does not intersect $\rightarrow$ keep.
    * S_3 intersects (at 3) $\rightarrow$ remove.
    * S_4 intersects (at 1 and 3) $\rightarrow$ remove.

    Reduced $\mathcal{S} = \{S_2\}$.
  - Recurse: $\textsc{EnumerateExactCovers}(U = \{2\}, \mathcal{S} = \{S_2\}, K = \{S_4\})$.
    * In recursion: U is not empty. Select $c = 2$ (only element, it appears in 1 set: S_2).
    * Iterate over sets containing $c = 2$: only S_2.
    * Try $R = S_2 = \{2\}$.
      · Add S_2 to K: $K = \{S_4, S_2\}$.
      · Let $V = \{2\}$.
      · Remove V from U: $U = \{2\} \setminus \{2\} = \emptyset$.
      · Remove from $\mathcal{S}$ all sets intersecting $V = \{2\}$:
        ◦ S_2 intersects (at 2) $\rightarrow$ remove.

        Reduced $\mathcal{S} = \emptyset$.

      - · Recurse: $\textsc{EnumerateExactCovers}(U = \emptyset,\ \mathcal{S} = \emptyset,\ K = \{S_4, S_2\})$.
        - ◦ U is empty $\to$ Output $K = \{S_4, S_2\}$ (covers $\{1, 2, 3\}$ exactly).
      - · Backtrack: Restore U to $\{2\}$ and $\mathcal{S}$ to $\{S_2\}$, remove S_2 from K: $K = \{S_4\}$.
    - ∗ No more sets containing $c = 2$. End recursion.
  - Backtrack: Restore U to $\{1, 2, 3\}$ and $\mathcal{S}$ to $\{S_1, S_2, S_3, S_4\}$, remove S_4 from K: $K = \emptyset$.

3. No more sets containing $c = 1$. Algorithm terminates.

Thus, the exact covers are $\{S_1, S_3\}$ and $\{S_4, S_2\}$.

The problem of enumerating all exact covers is NP-complete [8], implying that the theoretical complexity of Algorithm 1 is exponential. However, this NP-completeness is not easily exploitable to provide a cryptographic security in the suggested key exchange protocol over the lex-supertropical semiring, for the following reasons. First, only tangible equations are considered in the exact covering problem, and few such equations arise in transmitted messages of the protocol since many resolve to ghosts under the semiring operations. Second, the uniqueness of tangible equations coverings ensures that the subsets used to form the exact covers are mostly disjoint, resulting in very few exact covers overall. Third, the attack does not actually require enumerating all exact covers, it only requires finding one that satisfies the underlying linear system. In practice, such a cover can typically be found efficiently, often after testing only a few candidates (e.g., one or two).

We now present the process for obtaining a single solution to $A \otimes x = b$ over the lex-supertropical semiring through the two algorithms introduced below. Algorithm 2 computes a vector $\bar{x}_{\text{sup}}$, which serves as a foundation for constructing a solution to the system. The procedure involves evaluating f entrywise on A and b, finding the greatest solution $\bar{z}$ over $\mathbb{Z}^p_{\max}$, and either producing ghost entries or reconstructing tangible tuples for $\bar{x}_{sup}$. Here and below it is assumed that b has only finite entries and A has a finite entry in each column: we can make such assumption due to Remark 2.2.

Algorithm 2 Computing $\bar{x}_{\sup}$

Require: $A \in T^{m\times n}$, $b \in T^m$

Ensure: $\bar{x}_{\sup} \in T^n$ and $\widetilde{S}_j$ for $j \in [n]$

1: Compute $f(A)$ and $f(b)$ entrywise
2: TangibleEquations $\leftarrow \{\, i \in [m] : b_i$ is tangible $\}$
3: **for** $j = 1, \dots, n$ **do**
4:     For each i s.t. $A_{ij} \neq -\infty$ compute $d_{i,j} \leftarrow f(b_i) - f(A_{ij})$
5:     $\bar{z}_j \leftarrow \min\{d_{i,j} : A_{ij} \neq -\infty\}$
6:     $S_j \leftarrow \{\, i \in [m] : d_{i,j} = \bar{z}_j \,\}$
7:     $\widetilde{S}_j \leftarrow S_j \cap$ TangibleEquations
8:     **if** $\widetilde{S}_j = \emptyset$ **then**
9:         $\bar{x}_{\sup,j} \leftarrow \bar{z}_j$
10:     **else**
11:         **if** every A_{ij} is tangible for $i \in \widetilde{S}_j$ **then**
12:             Find a tangible $\bar{x}_{\sup,j}$ such that $A_{ij} + \bar{x}_{\sup,j} = b_i$ for some $i \in \widetilde{S}_j$.
13:         **else**
14:             $\bar{x}_{\sup,j} \leftarrow -\infty$
15: **return** $(\bar{x}_{\sup}, \{\widetilde{S}_j\}_{j=1}^{n})$

The following observation clarifies the connection between $\bar{x}_{\sup}$ and $\bar{z}$.

Proposition 3.1. We have $f(\bar{x}_{\sup,j}) = \bar{z}_j$ 1) for all j such that $\widetilde{S}_j = \emptyset$ and 2) for all j such that $\widetilde{S}_j \neq \emptyset$ and every A_{ij} is tangible for $i \in \widetilde{S}_j$.

Proof. For $\widetilde{S}_j = \emptyset$ this is immediate (see Algorithm 2 line 9), and in the second case using the linearity of f we obtain that $f(A_{ij}) + f(\bar{x}_{\sup,j}) = f(b_i)$ hence $f(\bar{x}_{\sup,j}) = f(b_i) - f(A_{ij}) = \bar{z}_j$ since $i \in \widetilde{S}_j$. $\square$

Note that $\bar{x}_{\sup}$ is not necessarily a solution. In particular, if there exist two distinct components j and j' such that $\widetilde{S}_j \cap \widetilde{S}_{j'} \neq \emptyset$, then at least one tangible equation will be violated, and consequently $\bar{x}_{\sup}$ is not a solution. Nonetheless, $\bar{x}_{\sup}$ serves as a basis for constructing a valid solution. Algorithm 3 extracts a valid solution x_{sol} from $\bar{x}_{\sup}$ by selecting an exact cover of the set of tangible equations via the sets $\widetilde{S}_j$.

Algorithm 3 Extracting a solution x_{sol} from $\bar{x}_{\sup}$

Require: $A \in T^{m\times n}$, $b \in T^m$, $\bar{x}_{\sup} \in T^n$, $\widetilde{S}_j$ for $j \in [n]$
Ensure: $x_{sol} \in T^n$

```
if A ⊗ x̄_sup = b then
    return x_sol ← x̄_sup
U ← { i ∈ [m] : b_i is tangible }
J ← { j : x̄_sup,j ≠ −∞ and S̃_j ≠ ∅ }
Sets ← { j ↦ S̃_j | j ∈ J }
for all exact covers K of U (e.g., using Algorithm 1) using Sets do
    x_sol ← x̄_sup
    for all j ∈ J \ K do
        x_sol,j ← −∞
    b′ ← A ⊗ x_sol
    if b′ = b then
        return x_sol
```

The complexity of Algorithm 2 is $O(mn)$, where m is the number of equations and n is the number of variables in the system, as it performs in $O(mn+m)$ entrywise applications of the map f, followed by per-column elementary operations (subtraction, minima, set collections) each taking $O(m)$ time. The complexity of Algorithm 3 is dominated by the enumeration of exact covers via Algorithm 1, plus $O(E \cdot mn)$ for verification over E covers, and non-enumeration parts (matrix-vector multiplications and set operations) contribute $O(mn)$.

The following proposition shows that the above described algorithms will always find a solution when the system is solvable.

Theorem 3.1. *Let $A \otimes x = b$ be a system over the lex-supertropical semiring such that each column of A has a finite entry and all entries of b are finite. If the system is solvable, then Algorithm 3 (with the help of Algorithm 2 and Algorithm 1) produces a solution x_{sol} satisfying*

$$A \otimes x_{sol} = b.$$

Proof. To prove that the algorithms always finds a valid solution, we consider any solution x to $A \otimes x = b$. The main idea of the following proof is that by modifying x we can construct a solution x_{sol} being one of the vectors found by Algorithm 3.

We first recall that

$$f(x_j) \leq \bar{z}_j \quad \forall j \tag{4}$$

by Corollary 2. Now let

$$K = \{j \mid x_j \text{ is tangible, } f(A_{ij}) + f(x_j) = f(b_i) \text{ for some tangible } b_i\}. \tag{5}$$

Note that for any such i we necessarily have that $f(A_{ij})+f(x_j)$ is the unique term among $f(A_{ik} \otimes x_k)$ that attains the maximum, in which case we also say that it uniquely dominates row i or that column j uniquely dominates row i (otherwise $\bigoplus_{j=1}^{n} A_{ij} \otimes x_j$ would produce a ghost), and we also have $A_{ij} + x_j = b_i$ with both A_{ij} and x_j being tangible.

Hence for each tangible b_i, there is an index $j \in K$ such that $f(A_{ij}) + f(x_j)$ uniquely dominates row i and is equal to $f(b_i)$ and, furthermore, $A_{ij} + x_j = b_i$ for this j.

Recall that we defined $\bar{z}_j = \min_l(f(b_l) - f(A_{lj}))$ for each j, where the minimum is taken over all l such that $A_{lj} \neq -\infty$. For rows i dominated by j, $f(x_j) = f(b_i) - f(A_{ij})$, so $\bar{z}_j \leq f(x_j)$. Combining this with (4) we obtain $\bar{z}_j = f(x_j)$ for all $j \in K$.

We now consider the following special cases, aiming to use x to construct appropriate x_{sol} as produced by Algorithm 3.

Case 1: $j \in K$. In this case we obtain $f(b_k) - f(A_{kj}) > f(x_j) = \bar{z}_j$ for all k such that $f(A_{kj}) + f(x_j) < f(b_k)$ and $f(b_k) - f(A_{kj}) = f(x_j) = \bar{z}_j$ for all k such that $f(A_{kj}) + f(x_j) = f(b_k)$. This shows that

$$\widetilde{S}_j = \{i : b_i \text{ tangible and } f(A_{ij}) + f(x_j) = f(b_i)\} \quad \forall j \in K \tag{6}$$

In particular, $\widetilde{S}_j \neq \emptyset$ for all $j \in K$ (by comparing (5) with (6)) and $A_{ij} + x_j = b_i$ for all $i \in \widetilde{S}_j$ with none of A_{ij} being ghosts implying that $x_j = \bar{x}_{\sup,j}$ using Algorithm 2 line 12. The collection $\{\widetilde{S}_j : j \in K\}$ partitions the tangible rows: each tangible equation i belongs to exactly one $\widetilde{S}_j$ (uniqueness). Hence, K is an exact cover of the set of tangible equations.

Case 2: $j \notin K$ but $\widetilde{S}_j \neq \emptyset$. In this case for some tangible row i we have $f(A_{ij}) + \bar{z}_j = f(b_i)$ while $f(A_{ij}) + f(x_j) < f(b_i)$ implying $f(x_j) < \bar{z}_j$. In this case we can assume $x_j = -\infty$ as $f(A_{ij} \otimes x_j) < f(b_i)$ for any $i \in [m]$ so that x_j can be set to the semiring zero.

Case 3: $j \notin K$ and $\widetilde{S}_j = \emptyset$. In this case we also have $f(x_j) \leq \bar{z}_j$ by (4). Replacing x_j by $\bar{x}_{\sup,j}$ does not violate any tangible equation since $\widetilde{S}_j = \emptyset$, and neither it violates any of the ghost equations.

We now see that solution x, possibly after some modifications described in the cases above, is precisely x_{sol} constructed in Algorithm 3 for some exact cover and therefore Algorithm 3 succeeds. □

4 Attacking the three-party key exchange protocol over the lex-supertropical semiring

The following protocol allows three parties to arrive at the same shared key because polynomials of the same matrix commute. Note that Ponmaheshkumar, Ramalingam and Perumal [10] used circulant matrices in the protocol. Here we analyze the polynomial version instead, since circulant matrices are just a special case of matrix polynomials.

Algorithm 4 Three-party key exchange protocol over the lex-supertropical semiring [10]

Require: Public matrices $A, B, W \in T^{n\times n}$
Ensure: Shared secret key K
1: Alice chooses private polynomials p_1, p_2 and computes $K_1 = p_1(A) \otimes W \otimes p_2(B)$, makes K_1 public
2: Bob chooses private polynomials q_1, q_2 and computes $K_2 = q_1(A) \otimes W \otimes q_2(B)$, makes K_2 public
3: Charlie chooses private polynomials r_1, r_2 and computes $K_3 = r_1(A) \otimes W \otimes r_2(B)$, makes K_3 public
4: Alice computes $K_4 = p_1(A) \otimes K_3 \otimes p_2(B)$ and sends K_4 to Bob
5: Bob computes $K_5 = q_1(A) \otimes K_1 \otimes q_2(B)$ and sends K_5 to Charlie
6: Charlie computes $K_6 = r_1(A) \otimes K_2 \otimes r_2(B)$ and sends K_6 to Alice
7: Alice computes $K_A = p_1(A) \otimes K_6 \otimes p_2(B) = p_1(A) \otimes (r_1(A) \otimes (q_1(A) \otimes W \otimes q_2(B)) \otimes r_2(B)) \otimes p_2(B)$
8: Bob computes $K_B = q_1(A) \otimes K_4 \otimes q_2(B) = q_1(A) \otimes (p_1(A) \otimes (r_1(A) \otimes W \otimes r_2(B)) \otimes p_2(B)) \otimes q_2(B)$
9: Charlie computes $K_C = r_1(A) \otimes K_5 \otimes r_2(B) = r_1(A) \otimes (q_1(A) \otimes (p_1(A) \otimes W \otimes p_2(B)) \otimes q_2(B)) \otimes r_2(B)$
10: **return** $K_A = K_B = K_C = K$

To attack this protocol, fix an integer D larger than the maximal degree that any party may use. Then, write matrices X and Y as:

$$X = \bigoplus_{\alpha=0}^{D} \left(x_\alpha \otimes A^{\otimes\alpha}\right), Y = \bigoplus_{\beta=0}^{D} \left(y_\beta \otimes B^{\otimes\beta}\right),$$

Then, K_1 can be expressed as:

$$K_1 = \bigoplus_{\alpha,\beta=0}^{D} x_\alpha \otimes y_\beta \otimes \left(A^{\otimes\alpha} \otimes W \otimes B^{\otimes\beta}\right).$$

Denote $R^{\alpha\beta} = A^{\otimes\alpha} \otimes W \otimes B^{\otimes\beta}$. For each entry (γ, δ) of K_1, we have:

$$\bigoplus_{\alpha,\beta=0}^{D} x_\alpha \otimes y_\beta \otimes \left(R^{\alpha\beta}\right)_{\gamma\delta} = (K_1)_{\gamma\delta} \quad \forall\gamma, \delta \in [n] \times [n]. \tag{7}$$

Introducing $z_{\alpha\beta} = x_\alpha \otimes y_\beta$, this becomes:

$$\bigoplus_{\alpha,\beta=0}^{D} z_{\alpha\beta} \otimes \left(R^{\alpha\beta}\right)_{\gamma\delta} = (K_1)_{\gamma\delta} \quad \forall\gamma, \delta \in [n] \times [n]. \tag{8}$$

Note that this is a system of linear equations of the shape $A \otimes x = b$ with coefficients $\left(R^{\alpha\beta}\right)_{\gamma\delta}$ and unknowns $z_{\alpha\beta}$. A solution to this system can be obtained using the approach outlined in the previous section.

For the shared secret key $K = p_1(A) \otimes K_6 \otimes p_2(B)$, where $K_6 = r_1(A) \otimes K_2 \otimes r_2(B)$, we recover K using a solution $(r_{\alpha\beta})$ to (8) as:

$$K = \bigoplus_{\alpha,\beta=0}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes K_6 \otimes B^{\otimes\beta}. \tag{9}$$

The formal attack algorithm is as follows, and let us mention that this is, essentially, a slight generalisation of the attack suggested by Otero-Sanchez et al. [9].

Algorithm 5 Attacking the three-party protocol based on (9)

Require: Public matrices A, B, W and K_6

Ensure: Shared secret key K

1: Find a solution $r_{\alpha\beta}$ of system (8) using Algorithm 3.

2: Compute the shared secret key K:

$$K = \bigoplus_{\alpha,\beta=0}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes K_6 \otimes B^{\otimes\beta}.$$

We now show that this attack recovers the actual shared secret key. From $K_6 = r_1(A) \otimes K_2 \otimes r_2(B)$ and $K_2 = q_1(A) \otimes W \otimes q_2(B)$, we get

$$\begin{aligned}
K &= \bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes K_6 \otimes B^{\otimes\beta} \\
&= \bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes \big(r_1(A) \otimes K_2 \otimes r_2(B)\big) \otimes B^{\otimes\beta} \\
&= \bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes r_1(A) \otimes A^{\otimes\alpha} \otimes K_2 \otimes B^{\otimes\beta} \otimes r_2(B) \\
&= r_1(A) \otimes \Big(\bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes K_2 \otimes B^{\otimes\beta} \Big) \otimes r_2(B).
\end{aligned}$$

Now substitute $K_2 = q_1(A) \otimes W \otimes q_2(B)$:

$$\begin{aligned}
K &= r_1(A) \otimes \Big(\bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes \big(q_1(A) \otimes W \otimes q_2(B)\big) \otimes B^{\otimes\beta} \Big) \otimes r_2(B) \\
&= r_1(A) \otimes q_1(A) \otimes \Big(\bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes W \otimes B^{\otimes\beta} \Big) \otimes q_2(B) \otimes r_2(B).
\end{aligned}$$

Because $(r_{\alpha\beta})$ is a solution of the K_1-system we have

$$\bigoplus_{\alpha,\beta}^{D} r_{\alpha\beta} \otimes A^{\otimes\alpha} \otimes W \otimes B^{\otimes\beta} \;=\; K_1 \;=\; p_1(A) \otimes W \otimes p_2(B).$$

Hence

$$K = r_1(A) \otimes q_1(A) \otimes K_1 \otimes q_2(B) \otimes r_2(B).$$

Substituting $K_1 = p_1(A) \otimes W \otimes p_2(B)$ gives

$$K = r_1(A) \otimes q_1(A) \otimes p_1(A) \otimes W \otimes p_2(B) \otimes q_2(B) \otimes r_2(B).$$

Finally, by commutativity of polynomial evaluations on the same matrix, we have

$$K = p_1(A) \otimes q_1(A) \otimes r_1(A) \otimes W \otimes p_2(B) \otimes q_2(B) \otimes r_2(B) = K_A = K_B = K_C.$$

Note that this attack is guaranteed to succeed since system (8) is solvable because it was originated by the protocol parties, and Theorem 3.1 ensures that Algorithm 3 will find a solution. This solution can then be substituted into the key-recovery formula, which, as shown above, successfully reconstructs the secret key.

Figure 1 shows the runtime of this attack implemented against the protocol using triples ($p = 3$) with f defined as in Example 2.1, compared with the execution of the protocol. Experiments use 10-dimensional matrices. The tuples of matrix entries and polynomial coefficients were sampled from the integer interval $[-1000, 1000]$. All reported times are averages over 10 trials. The attack attains a perfect success rate, as expected from the theoretical guarantee. We observe that the computational effort required by the attacker and by the honest parties remains comparable even for large polynomial degrees. These experiments were executed on Windows 11 64-bit, with an Intel(R) Core(TM) i7-9750H CPU @ 2.60GHz and 16.0 GB RAM.

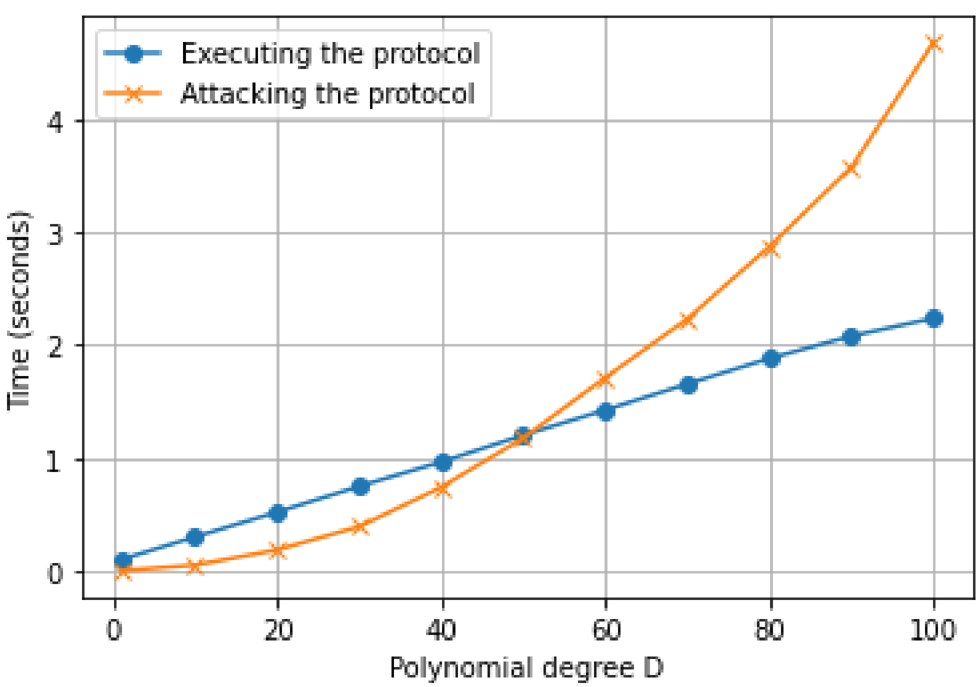

Figure 1: Executing vs. attacking Protocol 4

5 Conclusion

We have described a method for finding a solution of $A \otimes x = b$ over the lex-supertropical semiring, and used it to suggest an attack against a three-party key-exchange protocol put forward in [10]. The method combines a projection to the ghost part of T and an exact-cover enumeration to extract a valid solution to the system: Theorem 3.1 shows that this pipeline finds a solution. Casting the protocol's public messages as a linear system over the same semiring reduces the cryptanalytic problem to finding any valid solution to this system, from which Algorithm 5 deterministically recovers the shared secret key.

These findings suggest that this version of the modified supertropical semiring is not a secure platform for the considered multi-party protocol due to the easy solvability of the underlying linear systems. To mitigate this vulnerability, future protocol designs should avoid constructions that reduce to such easily solvable systems or consider more robust, layered semirings with higher layer complexity to prevent the easy solvability of these systems.

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A Proof of Proposition 2.1

Suppose z satisfies $A \otimes z = b$. Fix any column index j and any row i with $A_{ij} \neq -\infty$. From the equality

$$b_i = \max_k (A_{ik} + z_k),$$

we get, in particular, $A_{ij} + z_j \leq b_i$. Rearranging gives

$$z_j \leq b_i - A_{ij}.$$

This holds for every i with $A_{ij} \neq -\infty$, therefore

$$z_j \leq \min_{i: A_{ij} \neq -\infty} (b_i - A_{ij}) = \bar{z}_j.$$

Since this is true for every j, we have $z \leq \bar{z}$. Also, due to monotonicity, we have $A \otimes \bar{z} \geq A \otimes z = b$.

Conversely, fix a row i. For every column j with $A_{ij} \neq -\infty$ we have, by the definition of $\bar{z}_j$,

$$\bar{z}_j \leq b_i - A_{ij},$$

hence

$$A_{ij} + \bar{z}_j \leq b_i.$$

Taking the maximum over j yields

$$\max_j (A_{ij} + \bar{z}_j) \leq b_i,$$

i.e., the i-th component of $A \otimes \bar{z}$ satisfies $(A \otimes \bar{z})_i \leq b_i$. Since this holds for every i, we get $A \otimes \bar{z} \leq b$.

Combining the two inequalities gives $A \otimes \bar{z} = b$. Thus, whenever the system has a solution, it is satisfied by $\bar{z}$, and $\bar{z}$ is the greatest solution (any solution z satisfies $z \leq \bar{z}$). The implication in the other direction is trivial: if $A \otimes \bar{z} = b$, then the system is solvable.

B Proof of Proposition-Definition 2.1

- $(T, \oplus)$ is a commutative monoid.
  - Commutativity: Follows directly from Definition 2.5.
  - Identity: The element $-\infty$ satisfies $a \oplus -\infty = -\infty \oplus a = a$ for all $a \in T$, because $f(-\infty) = -\infty$ is minimal in the lexicographic order.
  - Associativity: We have $f((a \oplus b) \oplus c) = \max(\max(f(a), f(b)), f(c)) = \max(f(a), f(b), f(c)) = f(a \oplus (b \oplus c))$. If one projection is uniquely maximal, both sides yield the element with that projection (tangible or ghost). If two or more are equal maxima, both sides give the same ghost element. Cases involving $-\infty$ reduce to the identity property. Hence $(a \oplus b) \oplus c = a \oplus (b \oplus c)$.
- $(T, \otimes)$ is a monoid.
  - Identity: Let $e = (0, \ldots, 0) \in \mathbb{Z}^p$. Then, we have $a \otimes e = e \otimes a = a$ for all $a \in T$, in particular since $g(0, \ldots, 0) = (0, \ldots, 0)^\circ$. Thus e is the multiplicative identity.
  - Associativity: Since $f((a \otimes b) \otimes c) = f(a) + f(b) + f(c) = f(a \otimes (b \otimes c))$, both sides have the same projection $f(a) + f(b) + f(c)$. If no element is $-\infty$, they have the same layer (tangible if all elements are tangible, otherwise ghost). If any element is $-\infty$, both sides equal $-\infty$.
- $\otimes$ distributes over $\oplus$. Since $f(a \otimes (b \oplus c)) = f(a) + \max(f(b), f(c)) = \max(f(a) + f(b), f(a) + f(c)) = f((a \otimes b) \oplus (a \otimes c))$, both sides have the same projection. The case $a = -\infty$ is trivial. If $a \neq -\infty$ and $f(b) \neq f(c)$, the larger projection determines both sides. If $a \neq -\infty$ and $f(b) = f(c)$, then both sides yield the ghost $f(a) + f(b)$. Thus $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$.
- $-\infty$ is absorbing for $\otimes$. We have $a \otimes (-\infty) = (-\infty) \otimes a = -\infty$ for all $a \in T$.